Chambers, universal constructions, and realization cones

Human Compression project

Abstract

We organize the mathematical content of Schulte and Ivić Weiss's 2006 survey *Problems on Polytopes, Their Groups, and Realizations*. Three common arguments treat distinguished generators and closed flag walks, universal amalgamation with regular local types, and finite equivariant realizations. Two oriented amalgamations are verified by their complete presentations. Previously established group results answer the sporadic existence question and the regular alternating rank question within their precise scopes. The source's remaining classification inputs and questions are recorded separately. In particular, its two aggregate counts for locally projective polytopes are reported claims, not reproduced censuses. All four component manuscripts form part of the proof; this synthesis does not replace their proofs by numerical summaries.

1 The source and the mathematical interfaces

The fixed source is arXiv:math/0608397v1, dated 15 August 2006, corresponding to the paper in *Periodica Mathematica Hungarica* 53 (2006), 231–255. We use its twenty-five physical pages and its complete TeX source. A later compilation date on the acquired PDF does not constitute a later mathematical edition. This is a survey with thirty-four numbered problems, rather than a paper with a single new classification theorem. Fourteen additional records separate embedded conjectures, restrictions, and unnumbered research directions. They are not fourteen additional independent conjectures.

An abstract polytope of rank n is a ranked poset with unique least and greatest faces, the diamond condition, and strong flag connectedness. The diamond condition says that each section of rank two has exactly two intermediate faces. Strong flag connectedness requires a path between two flags that retains their common faces. These conditions concern actual flags and sections; they cannot be replaced by connectedness of an unrelated incidence graph.

For a regular polytope, its distinguished involutions $\rho_0, \ldots, \rho_{n-1}$ satisfy the string commutations and

$$\langle \rho_i : i \in I \rangle \cap \langle \rho_j : j \in J \rangle = \langle \rho_k : k \in I \cap J \rangle.$$

For a chiral polytope, adjacent flags are in the two different automorphism orbits. Its distinguished rotations $\sigma_1, \ldots, \sigma_{n-1}$ satisfy the string rotation relations and their rotation intersection condition. We retain the standard marked reconstruction theorems, attributed in the survey to McMullen–Schulte and Schulte–Weiss. Relations without the intersection condition do not supply a polytope. The geometric reconstruction is an explicit antecedent of the arguments below.

The complete general proof for highly symmetric hypertopes supplies the corresponding stabilizer argument. Paths within a residue identify its pointwise stabilizer with the subgroup on its remaining types. In the chiral case even paths give the pair-difference subgroup. Polytope flags are chambers of their face geometry, so the argument applies here. Conversely, a general C-group or C^+-group needs the geometric hypotheses stated in that proof. We use the marked string reconstruction where appropriate, not an unrestricted converse for arbitrary diagrams.

2 Words, orientations, and universal objects

Theorem 1. *For a regular or chiral polytope of rank $n \geq 3$, put $\alpha_0 = 1$ and $\alpha_i = \sigma_1 \cdots \sigma_i$ for $1 \leq i \leq n-1$. Simultaneous inversion of the α_i is equivalent to the usual reflection substitution*

$$\sigma_1 \longmapsto \sigma_1^{-1}, \qquad \sigma_2 \longmapsto \sigma_1^2 \sigma_2, \qquad \sigma_i \longmapsto \sigma_i \quad (i \geq 3).$$

For a chiral polyhedron, let s_1, s_2 denote its two distinguished rotations with the traversal convention in the component proof. The length of a j-hole is

$$\big|s_1 s_2^{1-j}\big|,$$

and the length of a j-zigzag is

$$2\big|s_1 s_2^{-j} s_1^{-1} s_2^{j}\big|.$$

Here $|g|$ denotes group order, and the traversal state includes the selected side of an edge.

Proof. The complete three-page component *Rotation conventions and closed flag walks* proves these statements by word cancellation and actual flag actions. The relevant identities are $\sigma_i = \alpha_{i-1}^{-1}\alpha_i$ and $\alpha_i^2 = 1$ for $i \geq 2$. If r_i is the operation of changing the type-i face of a flag, the traversal words are

$$h_j = r_0 r_1 (r_2 r_1)^{j-1}, \qquad z_j = r_0 (r_1 r_2)^j.$$

These r_i are flag permutations, not generally polytope automorphisms. An automorphism commutes with every adjacency operation. Consequently its iterates agree with successive repetitions of the chosen word on a base flag whenever one traversal returns to its automorphism orbit. The free action on that orbit identifies the traversal period with its group order. In a chiral action h_j preserves the orbit whereas z_j exchanges the two orbits. The component computes $h_j = s_1 s_2^{1-j}$ and $z_j^2 = s_1 s_2^{-j} s_1^{-1} s_2^j$ on the base flag. A return under z_j must therefore take an even number of steps, which gives the second formula. The same component gives the regular formulas and all convention changes explicitly. □

The formulas determine periods of a specified polyhedron. They do not determine which period lists determine a polyhedron, or when a universal quotient prescribed by those lists is finite. Those are the separate assertions in Problem 7.

Theorem 2. *Let K, L be regular rank-n polytopes, $n \geq 2$, with compatible distinguished common section. If any rank-$(n+1)$ polytope has facets K and vertex figures L, there is a regular universal polytope with these local types.*

Proof. The full two-page component *Universal regular amalgamation from a witness* gives the argument, including the group-action justifications. Start with involutions on the rank-$(n+1)$ string, imposing remote commutation. Add the complete marked relations of K on the first n generators and of L on the last n. In a given witness the adjacency permutations satisfy these relations at every flag: the local constituents are regular, so their marked relations hold throughout each local flag set. The presented group therefore acts on all witness flags.

The two end subgroups identify with the exact local groups. Surjections in one direction come from the local presentations; their actions on the corresponding local flags give the reverse order-free identifications. An element in both end subgroups preserves the base facet and vertex. Strong flag connectedness of their common section supplies a middle-subgroup element with the same effect on the base flag. The free action of either end group makes these two elements equal. Thus the end intersection is exactly the middle subgroup. The retained string end criterion and reconstruction theorem give a regular polytope. Repeating the action on every other witness supplies the covering map from this one, and proves universality. □

The witness assumption is essential. The source already gives a pair with no regular amalgamation: facets $\{4,4\}_{(3,0)}$ and hemicube vertex figures. That negative example is retained with its attribution to *Abstract Regular Polytopes*, p. 367. Theorem 2 supplies neither existence for arbitrary compatible pairs nor a finiteness criterion. The analogous theorem for oriented chiral local constituents is a separate retained Schulte–Weiss input. It requires the chosen enantiomorphic forms and directly regular constituents where a constituent is regular.

Proposition 3. *The two oriented universal amalgamations with local pairs*

$$(\{4,4\}_{(1,3)},\{4,4\}_{(1,3)}) \quad \textit{and} \quad (\{4,4\}_{(3,1)},\{4,4\}_{(1,3)})$$

are chiral and have full automorphism groups of orders 2000 *and* 960*, respectively.*

Proof. The complete two-page component *Two oriented square-torus amalgamations* verifies the full presentations

$$\langle u,v,w \mid u^4=v^4=w^4=(uv)^2=(vw)^2=(uvw)^2=1,\\ (u^{-1}v)^s(uv^{-1})^t=(v^{-1}w)(vw^{-1})^3=1\rangle$$

at $(s,t)=(1,3),(3,1)$. Exact isomorphisms from the finitely presented groups to permutation groups give the stated orders. The two end groups have order 40 and their intersection is $\langle v\rangle$ of order 4; both adjacent cyclic intersections are trivial. Separate local presentations also have order 40, so the local types do not collapse. Both toroidal relators fail after the reflection substitution of Theorem 1. The retained rotation reconstruction therefore gives chirality. The presentation imposes exactly the local relations, so its universal property is the retained oriented amalgamation correspondence. Independent literal permutation closure checks the finite orders and intersections; the upper order of the presentation itself remains the explicit GAP isomorphism calculation. These established examples are attributed by the survey to Hubard–Weiss (2005), p. 132. □

3 Finite equivariant realization theory

Let a finite group G act on a nonempty finite set X. A realization is a map $f: X \to V$ to a real orthogonal G-module with $f(gx)=gf(x)$. We allow repeated vertices until faithfulness is imposed, translate to centroid zero, and take the actual linear span of the image. Write $\mathbf{1}$ for the all-ones vector and $J=I-|X|^{-1}\mathbf{1}\mathbf{1}^T$.

Theorem 4. *Congruence classes of centered equivariant realizations are identified with*

$$\mathcal{C}_G(X)=\{B=B^T\succeq 0 : B\mathbf{1}=0,\ gB=Bg \text{ for every } g\in G\},$$

where g acts by its permutation matrix on $\mathbb{R}^X$. The dimension of the realized span is $\operatorname{rank} B$*. Squared distances and Gram matrices are mutually recoverable by*

$$D_{xy}=B_{xx}+B_{yy}-2B_{xy}, \qquad B=-\tfrac{1}{2}JDJ.$$

Blending realizations is addition in this cone. A nonzero B spans an extreme ray precisely when $\operatorname{im} B$ *is an irreducible real G-module.*

Proof. The complete four-page component *Finite realization cones* proves the Gram correspondence, the inverse on hollow symmetric distance matrices, and the extreme-ray statement. The realization of B is $f_x=B^{1/2}e_x$ in its actual image. Orthogonal direct sums give addition of Gram matrices; one then restricts to the span rather than assuming that the entire direct sum is spanned.

For clarity, the point where irreducibility enters is as follows. If $B = C + D$ with $C, D \succeq 0$, then $\ker B \subseteq \ker C$, so C restricts to $\operatorname{im} B$. An invariant self-adjoint operator on an irreducible real module is scalar: each of its real eigenspaces is invariant. This applies to both B and C, and makes C a scalar multiple of B. Conversely, if $\operatorname{im} B$ is reducible, choose an orthogonal invariant projection Q onto a nonzero proper summand. With $A = B^{1/2}$ the decomposition

$$B = AQA + A(I - Q)A$$

has two nonzero invariant positive semidefinite terms. Their ranks preclude either being a multiple of B. This argument does not assume that Q commutes with B. □

For a transitive vertex set $X = G/H$, all such orbit maps have the form $gH \mapsto gv$ with $v \in V^H$. Distinct abstract vertices have distinct images exactly when $\mathrm{Stab}_G(v) = H$. Faithfulness of the representation alone does not imply this equality. For a polytope one must also separate distinct faces of each rank under the recursive face construction. The component gives the precise recursive test, and identifies the full geometric symmetry group as the Gram stabilizer in the abstract automorphism group, acting on the actual affine span.

The finite-group assumption is used, in particular, in averaging and in the orthogonal representation setting. The source's infinite Euclidean Wythoff constructions, discreteness conditions, and geometric classifications are retained separately. Theorem 4 gives neither all individual toroid cones nor the answer about quaternionic irreducible characters in Problem 23. It also does not decide whether an abstract regular polytope has a faithful chiral geometric realization. The realization theory retains the attribution to McMullen and to McMullen–Monson recorded in the source; a new proof of its common linear algebra is not a claim of priority for that theory.

4 Group consequences already available

The acquired simple-group theorem of Leemans–Liebeck (2016) constructs a chiral polyhedron for every nonabelian finite simple group outside A_7, $\mathrm{PSL}_2(q)$, $\mathrm{PSL}_3(q)$, and $\mathrm{PSU}_3(q)$. We use its positive assertion only. All twenty-six sporadic groups lie in its scope, including the Monster, so it answers the existence request in Problem 31. This is a previously established answer, with its original attribution, not a new solution obtained here or an enumeration of all possible polytopes with those groups.

The acquired alternating-group existence and upper-rank theorems, together with their complete finite boundary censuses, give the following regular ranks for $n \geq 3$:

n	ranks of string C-markings generating A_n
$3, 4, 6, 7, 8$	none
5	3
9	$3, 4$
10	$3, 4, 5$
11	$3, 6$
$n \geq 12$	$3, 4, \ldots, \lfloor (n-1)/2 \rfloor$.

The exceptional gaps at $n = 11$ are retained from the complete finite census; they are not filled by interpolation. The lower existence theorem and upper bound have separate source attributions in their acquired manuscripts. The simple-group theorem also supplies chiral rank-three examples for A_n, $n \geq 8$. These statements do not classify all chiral ranks or all other polytopes requested in Problem 32.

For Problem 30, the acquired common quotient for a nondegenerate finite string C-marking of a 2-group in rank $d \geq 2$ has order 2^{2d-1} and nilpotency class at most two. The eight acquired families with large adjacent generator orders have their stated parameter hypotheses. Neither result classifies every 2-group, every chiral marking, or the groups of order $2^k p$ in the source question.

Proposition 5. *The full automorphism group of a finite neighborly regular or chiral polytope of positive rank acts* 2*-transitively on its vertices.*

Proof. First suppose the rank is at least three. Given an ordered pair (v, w) of distinct vertices, neighborliness supplies an edge e with these endpoints. Extend the incident vertex-edge pair (v, e) to a flag. In the regular case all flags lie in one orbit. In the chiral case, if the flag is in the unwanted orbit, change only its type-2 face. This preserves v and e and moves the flag into the chosen orbit. Do the same for any second ordered vertex pair (v', w'). An automorphism taking the resulting first flag to the second takes v to v' and e to e'. An edge has exactly two vertices by the diamond condition, so it also takes w to w'. Thus all ordered pairs of distinct vertices are in one orbit. In rank one the two vertices are interchanged. In rank two a neighborly polygon has two or three vertices, and its dihedral action is 2-transitive on them. Chiral polytopes have rank at least three by definition. □

This proves the necessary condition stated after Problem 33. It is not a classification or a converse, and is not claimed as a new theorem in the subject.

5 Retained classifications and precise limits

The survey collects results from many different sources. The accompanying register has fifty-one clauses in sixteen contexts, with source pages, literal anchors, citation keys, and each acquired proof or retained premise. Its sixty-six bibliography entries preserve the complete source list. The following table describes the division of work. A retained classification is an attributed input, not a classification reproved by the arguments above.

Context and register	Proof or retained input
C01: Definitions and chambers R01	Adopt the definitions and retain the marked reconstruction theorems. The acquired residue proof supplies the stabilizer identities under actual polytope hypotheses.
C02: Orientations R02	The complete word proof identifies both reflection conventions. Index at most two for rotations does not mean index exactly two.
C03: Dualities and maps R03–R08	Retain the Archdeacon–Richter and Servatius sphere results, Hubard–Weiss polarity theorem, Jendrol counterexample, and attributed toroidal and genus statements. The extra fully transitive toroidal family is not chiral.
C04: Hole and zigzag periods R09	The full flag-walk proof gives all stated periods. It gives no classification from period data.
C05: Amalgamation R10–R16	Prove the regular universal theorem from a witness and reproduce the two oriented examples. Retain oriented universality, the ARP cross-polytope criterion, the negative pair, and historical extension/rank-five assertions.
C06: Global topology R17–R19	Retain the spherical, projective and toroidal classifications and chiral restrictions from Coxeter and ARP, with their tessellation hypotheses.
C07: Local spherical and projective types R20–R23	Retain the universal Coxeter tessellation theorem. Acquire the exact J1 times PSL2(19) example and its corrected family census. The global17/441 aggregates remain unverified source reports.
C08: Local toroidal types R24–R29	Retain the precise rank/type and regular finite-family classifications in ARP. Conditional regular universality explains the rank-five/six universal restriction. Finite-list conjectures remain separate.
C09: Other local surfaces R30–R31	The four orientability combinations are definitions. Retain the historical catalogue report with its unverified chiral-citation scope.
C10: Equivariant realizations R32,R50	Prove the finite orbit-stabilizer and recursive faithfulness criteria. Infinite Euclidean Wythoff constructions remain separate retained inputs with discreteness and exact stabilizers required.

Context and register	Proof or retained input
C11: Realization cones	Prove the common centered Gram-cone and irreducibility theorems.
R33–R34	Individual cone catalogues and unevaluated character identities retain their published scope.
C12: Geometric classifications	Retain Grunbaum–Dress, Schulte, McMullen, and the finite planar-faced uniform classifications. Skew-faced and infinite cases are not covered by the
R35–R38	finite planar list.
C13: Medial graphs	Retain the precise rank-four type{3,q,3} results and their arc-transitivity or
R39–R40	semisymmetry hypotheses. The reverse construction has a separate polytopality condition.
C14: Petrie schemes	Retain the attributed positive geometric classes and necessary local
R41–R42	condition. A return of a full flag does not exclude repetition of a proper face.
C15: Specified groups	Acquire the complete sporadic existence and regular alternating ranks, and
R43–R46,R51	only the specified restricted2-group results. Prove the neighborliness implication, with no classification converse.
C16: Quotients	Retain Hartley quotient and semisparseness theorems with every conjugate
R47–R49	quantifier. The general end-condition converse remains a distinct question.

There are two numerical limits of particular importance. The source reports seventeen universal locally projective regular 4-polytopes, eight with a 2 in their Schläfli symbol, and 441 quotients including four additional nonuniversal regular polytopes. These are attributed to Hartley's then-preprint classification. We have not reproduced the global lists or reconciled the complete preprint and published editions. They therefore remain reported aggregates, not verified conclusions of this paper.

One constituent is acquired in full: the universal $\{\{5,3\},\{3,5\}_5\}$ has group $J_1 \times \mathrm{PSL}_2(19)$, and its accepted corrected quotient census has 137 entries under its specified equivalence. It would be invalid to subtract a correction for this family from 441 without the original global partition, overlap conventions, and edition correspondence. No corrected global total is asserted.

The source's report of map enumerations through orientable genus 15 and nonorientable genus 30 includes a chiral claim whose cited paper is titled *Determination of all regular maps of small genus*. The inspected material does not independently supply that chiral census. We retain the historical report with this explicit limitation.

The source's semisparseness criterion is likewise retained with its actual quantifiers. Put $\Gamma_{<i} = \langle \rho_j : j < i \rangle$, $\Gamma_{>i} = \langle \rho_j : j > i \rangle$ and $\Gamma_i = \langle \rho_j : j \neq i \rangle$. For $\Sigma \leq \Gamma$, the first two tests are

$$\Sigma^\varphi \cap \Gamma_i \rho_i = \varnothing,$$
$$\Gamma_{>i} \cap \Gamma_{<j} \Sigma^\varphi \subseteq \Gamma_k (\Gamma_{>i} \cap \Gamma_{<j}) \Sigma^\varphi, \qquad i < k < j,$$

for every $\varphi \in \Gamma$ and all displayed valid indices. The third is written in the source as

$$\Gamma_{>i} \cap \Gamma_{<i+1} \Sigma \subseteq \Sigma \quad (0 \leq i < n-1).$$

Hartley's further necessary condition requires both end intersections to be semisparse for every conjugate and

$$\Sigma^\varphi \cap \Gamma_0 \Gamma_{n-1} = (\Sigma^\varphi \cap \Gamma_0)(\Sigma^\varphi \cap \Gamma_{n-1}) \quad (\varphi \in \Gamma).$$

Problem 34 asks for sufficiency of these latter conditions. The special locally projective census, with its additional hypotheses, does not prove that general converse.

6 The questions and the fixed edition

The table below records every numbered problem. "Unsettled here" states a limit of the present acquired proofs; it does not assert that the question is still open in the current literature. A

broader search for later answers is outside this intake. The fourteen secondary records follow the table so that an embedded conjecture cannot disappear within a broader programme.

No.	Question in the fixed edition	Disposition and qualification
1	Classify self-dual toroidal polyhedra.	Unsettled here. The spherical classification is attributed background; regular or chiral toroidal maps are only a subclass.
2	Decide whether every equivelar self-dual polyhedron has a polarity.	Unsettled here. The source gives the affirmative regular/chiral subclass and a negative unrestricted self-duality example; neither answers the equivelar question.
3	Classify fully transitive polyhedra on surfaces of small genus, with both orientability choices.	Unsettled here. The toroidal classification cited in the source includes a further family whose middle adjacency preserves each flag orbit.
4	Determine the least orientable genus at least two admitting a nonregular fully transitive polyhedron.	Unsettled here. The source excludes chiral maps in genera two through six; fully transitive is a larger class.
5	Characterize groups of fully transitive polyhedra outside the regular and chiral classes.	Unsettled here. The subsequent self-dual restriction is an additional suggested subproblem.
6	Classify edge-transitive polyhedra on orientable and nonorientable surfaces of small genus.	Unsettled here. The cited partial results do not give the requested classification.
7	Determine what all specified hole and zigzag lengths determine, and when the associated universal regular quotient is finite.	Unsettled here. The first question covers regular or chiral polyhedra of type {p,q}, with 1¡=j¡=floor(q/2). The universal regular quotient in the subsequent question is distinct. Word-order formulas alone give neither uniqueness nor a universal finiteness criterion.
8	Decide existence with prescribed regular or chiral facets and vertex figures.	Already negative in the source. This is already negative in general in the source: the cited toroidal (3,0) facet and hemicube vertex figure have no regular amalgamation.
9	Characterize the compatible facet and vertex-figure pairs for which amalgamation exists.	Unsettled here. Matching their common sections is necessary but insufficient.
10	Determine universal groups and their finiteness for regular facet and vertex-figure pairs admitting amalgamation.	Unsettled here. The source existence theorem for the universal object is retained separately from group classification.
11	Determine universal groups and finiteness for oriented chiral/regular amalgamations with at least one chiral member.	Unsettled here. Choose enantiomorphic forms, with at least one chiral constituent and every regular constituent directly regular. Simultaneous reversal identifies four oriented choices in pairs; if one constituent is regular the remaining choices also agree. Existence remains a prerequisite.
12	Determine existence and finiteness with an arbitrary finite regular facet and a convex regular vertex figure.	Unsettled here. The source records the cross-polytope vertex-figure case: finiteness is equivalent to neighborliness of the facet.
13	Determine existence and finiteness with finite chiral facets and convex regular vertex figures.	Unsettled here. The regular cross-polytope result does not automatically extend to chiral facets.
14	Establish existence of finite chiral polytopes in ranks at least six.	Unsettled here. Affirmative answer conjectured in the 2006 source. This inventory makes no present-day open-status assertion.
15	Decide whether a convex four-polytope can have every facet combinatorially icosahedral.	Unsettled here. An abstract polytope or incidence geometry with such facets is not sufficient.

No.	Question in the fixed edition	Disposition and qualification
16	Classify universal regular amalgamations of square toroidal maps with odd distinct parameters s,u at least three.	Unsettled here. Two claims are embedded: existence for every such pair and finiteness only for (3,5) and (5,3).
17	Classify universal regular amalgamations of triangular and hexagonal toroidal maps in the stated regular parameter families.	Unsettled here. Retain the exact source parameter domains: s¿=2,t=0 or s=t¿=1, and the corresponding u,v alternatives.
18	Complete the classification of the three locally toroidal universal regular rank-six families, up to duality.	Unsettled here. The completeness of each finite list and the explicit one-relation infinite-family conjecture are separate clauses.
19	Classify the universal locally toroidal chiral rank-four polytopes with the specified regular or chiral constituents and enantiomorphic forms.	Unsettled here. The preceding discussion and subsequent group calculation focus on the finite classification; retain that goal separately from the literal universal-object wording. At least one constituent is a chiral toroidal map. The square-torus/cube subfamily has its own four-pair finiteness conjecture.
20	Classify finite universal regular or chiral rank-four polytopes with prescribed small facet and vertex-figure genera and orientabilities.	Unsettled here. Duality exchanges the two mixed orientability cases; the problem is not restricted to locally toroidal polytopes.
21	Develop the realization theory of chiral polytopes.	Unsettled here. A theory of equivariant Gram matrices alone does not classify faithful realizations or determine their full geometric symmetry.
22	Describe realization cones of regular toroids in every rank.	Unsettled here. The cited rank-three cones are established background, not the requested all-rank description.
23	Decide whether a finite string C-group can have a complex irreducible character with Frobenius-Schur indicator minus one.	Unsettled here. The representation must be quaternionic; a general real representation or a non-C-group example is insufficient.
24	Determine intersection-property validity from character tables or other readily computed group-algebra invariants for a marked string-generated group.	Unsettled here. The marking is part of the input. An explicit subgroup enumeration is not by itself an answer to the character-data question.
25	Enumerate discrete fully transitive polyhedra in Euclidean three-space, both finite and infinite.	Unsettled here. Regular and chiral classifications supply subclasses only.
26	Enumerate discrete uniform polyhedra in Euclidean three-space, both finite and infinite.	Unsettled here. The finite planar-faced list does not include all skew-faced or infinite cases.
27	Characterize when the source construction from a finite bipartite trivalent two- or three-arc-transitive graph yields a polytope.	Unsettled here. The condition must concern the actual reconstructed poset; graph transitivity alone is not asserted sufficient.
28	Decide whether all Petrie schemes are acoptic for each of four specified classes of products, uniform convex polytopes and regular star-polytopes.	Unsettled here. The four classes are: products of cubes and crosspolytopes; products of cubes or crosspolytopes and 4k-gons (k¿=1); uniform convex polytopes; regular star-polytopes. Affirmative answers are conjectured for classes 1,2,4; class 3 remains a separate question.
29	Characterize universal regular amalgamations for which every Petrie scheme is acoptic.	Unsettled here. The stated local nonintersection condition on constituents is only necessary.

No.	Question in the fixed edition	Disposition and qualification
30	Characterize full automorphism groups of regular or chiral polytopes of order a power of two or a power of two times an odd prime.	Restricted acquired results only; general question unsettled here. Existing project results cover specified ordered regular markings, bounded class/excess and exact small excesses; the unmarked, chiral and mixed-prime problem remains outside them.
31	Find polytopes with sporadic simple full automorphism groups, explicitly including the Monster.	Affirmative existence by the acquired 2016 theorem. The accepted positive theorem of the 2016 simple-group manuscript gives a chiral polyhedron for each of the26 sporadic groups. This is an acquired earlier answer, not a new conjecture proof.
32	Find regular, chiral, or other polytopes with full automorphism group A_n; in particular determine the possible degrees for each rank in the regular or chiral classes.	Regular ranks acquired; chiral rank-three existence only; other parts unsettled here. For n¿=3, acquired L038/L044 and their complete small censuses settle the regular ranks: A5:{3}; A9:{3,4}; A10:{3,4,5}; A11:{3,6}; none at n=3,4,6,7,8; and exactly 3¡=r¡=floor((n-1)/2) for n¿=12. Acquired L045 gives chiral rank-three existence for n¿=8. This does not classify all chiral ranks or all other polytopes. These are attributed earlier results, not newly solved questions.
33	Classify finite neighborly regular or chiral polytopes.	Unsettled here. The necessary two-transitive vertex action is not asserted sufficient for reconstruction.
34	Decide whether semisparsity follows from semisparsity of both end intersections in every conjugate and the displayed product-intersection equality.	Unsettled here. Retain every conjugate quantifier; necessity and the general three-condition definition do not prove this sufficiency assertion.

Secondary records.

Record	Exact scope of the additional record
Problem 5: selfdual	Seek the group characterization first for self-dual fully transitive polyhedra. No new solution supplied.
Problem 14: existence	Affirmative finite chiral existence conjecture in every rank at least six. No new solution supplied.
Problem 16: existence	Existence for all odd distinct s,u¿=3. No new solution supplied.
Problem 16: finiteness	Finite precisely at (3,5) or its dual. No new solution supplied.
Problem 18: lists	The known finite lists in all three rank-six families are complete. No new solution supplied.
Problem 18: first family	The specified Coxeter quotient is an infinite C-group outside (2,0,0,0),(2,2,0,0),(3,0,0,0). No new solution supplied.
Problem 19: cube	The square-torus/cube universal chiral group is finite precisely at (1,2),(1,3),(1,4),(2,3), up to transposition. No new solution supplied.
faithful chiral	Can an abstract regular polytope admit a faithful chiral geometric realization? No new solution supplied.
abstract uniform	Define abstract uniform polytopes and develop their abstract and geometric theories. No new solution supplied.
helices	Determine general properties of polyhedra with helical faces. No new solution supplied.
Problem 28: three classes	All Petrie schemes are acoptic in classes1,2,4. No new solution supplied.

Record	Exact scope of the additional record
Problem 31: finite simple	Corresponding existence question for arbitrary finite simple groups, with the historical regular-map question distinguished. Only the exact positive families of the acquired simple-group theorem are supplied; the excluded families and regular-map question are not decided.
star vertex figures	Extend the existence/finiteness questions by allowing a regular star-polytope as vertex-figure. No new solution supplied.
prescribed facets	Find and describe regular or chiral polytopes with a prescribed facet isomorphism type, including variants with a prescribed last Schlaefli entry. No new solution supplied.

Several edition qualifications matter for interpreting these statements. Within the nondegenerate fundamental domain used in the source, the chiral toroid condition is $t \neq 0$ *and* $s \neq t$; the printed "or" would include the regular cases. This correction alone does not prove that every small parameter gives a nondegenerate polytope. The reference to plane maps at rank $n = 2$ is inconsistent with the source's rank-three polyhedron convention and is retained as an edition qualification. Recursive face realization is used at all appropriate ranks, rather than extrapolating the displayed $j = 1, 2, 3$ notation without a definition. The abstract and introduction give different meeting years; this historical discrepancy does not change the fixed mathematical edition. Three repeated TeX labels refer to different numbered problems, so our register identifies problems by ordinal number and source position, never by the duplicated label alone.

7 Proof record and attribution

The three common theoretical components contain complete proofs: four pages for finite realization cones, three for rotation conventions and flag walks, and two for conditional regular amalgamation. The two finite examples add two pages with complete presentations, executable GAP calculations, and independent literal permutation checks. All four have distinct reviews. Their unabridged texts are appended to the reading volume. The earlier group and locally projective results are acquired with their full proof and certificate dependencies, not just their final orders or page counts.

The new work here is the arrangement of these proofs, the explicit joins between their hypotheses, and the short proof of Proposition 5. The source authors and their cited predecessors retain attribution for established statements. The source register separately identifies definitions, retained classifications, historical reports, acquired proofs, and questions. No new open-problem solution is claimed. The computational checks are not Lean certificates, and no new compression factor is inferred from this survey disposition. Its admission as a handled original requires the separate review of this entire assembly.

References

[1] E. Schulte and A. Ivić Weiss, *Problems on Polytopes, Their Groups, and Realizations*, Period. Math. Hungar. 53 (2006), 231–255; arXiv:math/0608397v1. Fixed source and complete bibliography.

[2] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002. Standard reconstruction and the classifications at the locations specified in the retained-input register.

[3] E. Schulte and A. Ivić Weiss, *Chirality and projective linear groups*, Discrete Math. 131 (1994), 221–261. Retained through the inspected survey, not independently reacquired here.

[4] I. Hubard and A. Ivić Weiss, *Self-duality of chiral polytopes*, J. Combin. Theory Ser. A 111 (2005), 128–136. Attribution and oriented examples as cited in the fixed source.

[5] Human Compression project, *Finite realization cones*; *Rotation conventions and closed flag walks*; *Universal regular amalgamation from a witness*; *Two oriented square-torus amalgamations*. Complete acquired manuscripts and distinct reviews.

[6] The exact acquired group, chamber, and locally projective manuscripts are listed with their source attributions and immutable editions in the acquired-proof register. Their original results receive no new priority or question credit in this paper.

Realization cones from a finite vertex action

Human Compression project

The realization theory discussed in Section 6 of Schulte–Weiss [1] has a common linear algebraic description. Once the action on vertices is fixed, its realizations are described by positive semidefinite matrices commuting with that action. We give the argument, including the distinction between an equivariant placement, a faithful realization of faces, and a realization with a prescribed full symmetry group. The cone and its representation-theoretic interpretation are established parts of realization theory, attributed in the survey to McMullen and McMullen–Monson. They are not new theorems claimed by this project.

1 Distances and centered Gram matrices

Let X be a finite nonempty set of cardinality m, and let a finite group G act on X. Write $M = \mathbb{R}^X$ with its usual orthonormal basis $(e_x)_{x\in X}$, and let $P_g e_x = e_{gx}$. Put

$$J = I - \frac{1}{m}\mathbf{1}\mathbf{1}^{\mathsf{T}}, \qquad M_0 = \mathbf{1}^{\perp}.$$

A placement is a family $f = (f_x)_{x\in X}$ of points in a Euclidean space. We always replace the ambient space by the affine span of these points, and identify placements differing by an isometry that preserves the labels. Subtracting the centroid makes $\sum_x f_x = 0$. The affine span is then the linear span, denoted W.

Theorem 1. *The centered placements for which every $g \in G$ is induced by an isometry are classified, up to labeled congruence, by the cone*

$$\mathcal{C}_G = \{B \in \mathbb{R}^{X\times X} : B = B^{\mathsf{T}},\ B \succeq 0,\ B\mathbf{1} = 0,\ P_g^{\mathsf{T}} B P_g = B \text{ for every } g \in G\}.$$

The dimension of the placement is $\operatorname{rank} B$*. Its squared distance matrix is*

$$D_{xy} = B_{xx} + B_{yy} - 2B_{xy}, \qquad B = -\tfrac{1}{2}JDJ. \tag{1}$$

These formulas give mutually inverse linear maps between $\mathcal{C}_G$ and the invariant squared distance matrices.

Proof. For a placement define $T : M \to W$ by $Te_x = f_x$. Its Gram matrix is $B = T^*T$. It is positive semidefinite and symmetric, and $T\mathbf{1} = 0$ gives $B\mathbf{1} = 0$. An isometry permuting the points preserves their centroid, so after centering it is orthogonal. Its preservation of inner products is exactly the displayed invariance of B.

Conversely let $B \in \mathcal{C}_G$. The positive semidefinite square root $B^{1/2}$ commutes with every P_g, since each eigenspace of B is G-invariant. Set $W = \operatorname{im} B$ and $f_x = B^{1/2}e_x$. These vectors span W, their sum is zero, and g acts on them by the orthogonal restriction of P_g. Their Gram matrix is B.

If two spanning families f_x and f'_x have this Gram matrix, the rule $\sum_x a_x f_x \mapsto \sum_x a_x f'_x$ is well-defined: the squared norm of either sum is $a^{\mathsf{T}}Ba$. It is an onto linear isometry. Thus the matrix determines the labeled placement, including degenerate placements. Its rank is the dimension of the span.

The first formula in (1) follows by expanding $\|f_x - f_y\|^2$. Multiplying on both sides by J annihilates the two terms constant in one coordinate, and $JBJ = B$, which proves the second formula. To see that no additional condition on a proposed distance matrix has been hidden, take any symmetric hollow matrix D and set $B = -JDJ/2$. If $r_x = m^{-1}\sum_y D_{xy}$ and $s = m^{-2}\sum_{x,y} D_{xy}$, then

$$B_{xy} = -\tfrac{1}{2}(D_{xy} - r_x - r_y + s), \qquad B_{xx} = r_x - s/2.$$

Consequently $B_{xx} + B_{yy} - 2B_{xy} = D_{xy}$. If $B \succeq 0$, the construction already given realizes D. Invariance passes in both directions because J commutes with permutation matrices. □

In particular, one may specify D by one coordinate for each G-orbit on unordered pairs of distinct vertices. These are the diagonal classes used in the survey. The conditions in Theorem 1 show directly that the set of possible diagonal vectors is a closed convex cone: it is linearly isomorphic to the intersection of the positive semidefinite cone with a linear subspace. No assertion that all of its points are faithful is involved.

If f and h are two centered placements, their blend is the placement $x \mapsto (f_x, h_x)$ in $W_f \oplus W_h$, restricted to the span of these vectors. Its Gram matrix is $B_f + B_h$, and its squared distance matrix is $D_f + D_h$. The restriction to the actual span matters when the two representations contain equivalent summands. Multiplication of the matrices by $c \geq 0$ corresponds to multiplication of the placement by $\sqrt{c}$.

2 Irreducible realizations and fixed vectors

Theorem 2. *A nonzero $B \in \mathcal{C}_G$ spans an extreme ray of $\mathcal{C}_G$ if and only if the real G-module $\operatorname{im} B$ is irreducible. Every element of the cone is a sum of matrices on extreme rays.*

Proof. First suppose $W = \operatorname{im} B$ is irreducible and $B = C + D$ with $C, D \in \mathcal{C}_G$. If $v \in \ker B$, positivity gives $0 = \langle Cv, v\rangle + \langle Dv, v\rangle$, so both terms vanish. Diagonalizing a positive semidefinite matrix shows that a zero value of its quadratic form puts v in its kernel. Thus C and D vanish on $\ker B$, and symmetry makes them preserve $W = (\ker B)^{\perp}$. Every eigenspace of the symmetric operator $C|_W$ is G-invariant. Irreducibility therefore makes $C|_W$ a scalar operator. The same holds for $B|_W$, whose scalar is positive. It follows that C is a nonnegative multiple of B; the same is true of D. Hence the ray is extreme.

Conversely suppose W has a nonzero proper invariant subspace U. Its orthogonal complement is invariant, since the action is orthogonal. Let Q be orthogonal projection onto U, extended by zero on $W^{\perp}$. Then Q commutes with G. With $A = B^{1/2}$, the decomposition

$$B = AQA + A(I - Q)A$$

consists of two members of $\mathcal{C}_G$. Both are nonzero, since A is invertible on W, and their ranks are respectively $\dim U$ and $\dim W - \dim U$. Neither is proportional to B. Thus its ray is not extreme. Notice that B need not commute with an arbitrarily chosen invariant projection; the square roots on both sides of Q avoid that unjustified assumption.

For the final assertion use the placement f_x constructed in Theorem 1. Its orthogonal representation on W splits into a finite orthogonal sum of irreducible invariant subspaces: choose a minimal nonzero invariant subspace, take its orthogonal complement, and continue by induction on dimension. Projecting every f_x onto each summand gives centered equivariant placements. Their Gram matrices sum to B, and each nonzero one has that irreducible summand as its span, because the original f_x span W. The first assertion applies. □

Now assume that the action on X is transitive. Choose $x_0 \in X$ and put $H = \operatorname{Stab}_G(x_0)$, so that X is identified with the left cosets G/H. For an orthogonal representation $\rho : G \to O(V)$ write $V^H = \{v : \rho(h)v = v \text{ for every } h \in H\}$.

Proposition 3. *Equivariant placements of G/H in V are precisely*

$$f_{gH} = \rho(g)v \qquad (v \in V^H).$$

The placement is injective exactly when $\mathrm{Stab}_G(v) = H$. *Its centroid is the projection of v onto V^G. If V is nontrivial irreducible and $0 \neq v \in V^H$, the placement is centered, spans V, and gives an extreme ray of $\mathcal{C}_G$.*

Proof. The displayed expression is independent of the representative of gH exactly when v is fixed by H. Equivariance forces every placement to have this form, by taking $v = f_H$. Two cosets have the same image precisely when their quotient lies in $\mathrm{Stab}_G(v)$, which proves the injectivity assertion.

The mean of the orbit is $|G|^{-1}\sum_{g\in G}\rho(g)v$: every coset appears $|H|$ times in this sum. The averaging operator is self-adjoint and idempotent, and has image V^G, hence is the orthogonal projection onto that subspace. For a nontrivial irreducible representation $V^G = 0$, and the invariant span of the orbit of a nonzero vector is V. The last assertion follows from Theorem 2. □

This identifies the Wythoff space with a fixed-vector space. Different irreducible representations need not contribute, since V^H may be zero. Within a representation, different fixed vectors need not give the same ray. Thus an enumeration of irreducible characters alone is not a description of the realization cone. No assumption about the real, complex or quaternionic type of an irreducible representation was used. In particular, this argument does not decide Problem 23 of the survey.

3 Faces and full geometric symmetry

Let $\mathcal{P}$ be a finite abstract polytope with vertex set X. A vertex placement induces the recursive face images used in the survey: the image of a face of rank $j \geq 1$ is the set of images of its incident faces of rank $j-1$. Consider the following condition on the abstract polytope:

$$F \neq F' \text{ of rank } j \quad \Longrightarrow \quad \{E : \operatorname{rank} E = j-1,\ E < F\} \neq \{E : \operatorname{rank} E = j-1,\ E < F'\}. \quad (2)$$

It is required for every $j \geq 1$ for which there are faces to compare.

Proposition 4. *The recursive realization is faithful on every rank if and only if the vertex placement is injective and* (2) *holds. Subject to this combinatorial condition, faithfulness for $B \in \mathcal{C}_G$ is exactly $D_{xy} > 0$ for distinct vertices x, y.*

Proof. If two distinct faces have the same set of incident faces of the preceding rank, their images agree for every vertex placement. Faithfulness therefore requires (2), as well as injectivity on vertices. Conversely, suppose the images in rank $j-1$ are distinct. Equality of two sets of such images then implies equality of the two underlying sets of faces. Condition (2) makes the rank-j images distinct. Induction from rank zero proves the assertion. Finally $D_{xy} = \|f_x - f_y\|^2$, so its strict positivity is exactly vertex injectivity. □

Suppose (2) holds and the placement is faithful. Put $A = \mathrm{Aut}(\mathcal{P})$, acting on X. This action is faithful: an automorphism fixing every vertex fixes each succeeding rank by (2). The realized faces retain their ranks and incidence.

Proposition 5. *The full symmetry group of the faithful realization, in its affine span, is naturally isomorphic to*

$$A_B = \{a \in A : P_a^{\mathsf{T}} B P_a = B\}.$$

Consequently, for $G \leq A$, a faithful equivariant placement has full geometric group exactly G if and only if $B \in \mathcal{C}_G$ and no $a \in A \setminus G$ preserves B.

Proof. A symmetry of the realization permutes its faces preserving ranks and incidence. Faithfulness gives an automorphism of $\mathcal{P}$, and the symmetry preserves all squared distances. By (1) it preserves B. Conversely, if $a \in A$ preserves B, the same argument with linear relations used in Theorem 1 shows that $f_x \mapsto f_{ax}$ extends to an orthogonal isometry of the centered span. The recursive definition of face images makes it act on every face as a. Both constructions are unique because the vertices span. They preserve composition and are inverse to one another. □

The distinction in Proposition 5 is necessary for chiral realizations. A group representation and a choice of initial vertex may have more symmetry than the designated group. The proposition supplies an exact criterion for that issue for a fixed finite polytope; it asserts no existence or classification result for all polytopes.

4 Use in the source disposition

Theorem 1 proves the finite cone and diagonal-vector statements in the survey, and identifies blending with addition. Theorem 2 proves the statement about pure realizations and irreducible orthogonal representations. Proposition 3 supplies the fixed-space interpretation, with the stabilizer condition that distinguishes an injective orbit from a merely faithful representation. Propositions 4 and 5 state the further obligations needed for faithful faces and the full geometric group.

These arguments do not compute the cones for every regular toroid, classify discrete realizations in dimension three, decide the quaternionic question, or exhibit a faithful chiral realization of a regular abstract polytope. They do not replace the separately attributed geometric classifications in the survey. The present manuscript rederives common established theory; there is no new conjecture, Lean or compression-measurement credit.

References

[1] E. Schulte and A. Ivić Weiss, *Problems on polytopes, their groups, and realizations*, Period. Math. Hungar. 53 (2006), 231–255; fixed arXiv math/0608397v1, Section 6 and its references to P. McMullen, *Realizations of regular polytopes*, Aequationes Math. 37 (1989), 38–56, and P. McMullen and B. R. Monson, *Realizations of regular polytopes, II*, Aequationes Math. 65 (2003), 102–112. Attribution here follows the read survey; its cited classifications are not re-established by the matrix arguments above.

Rotation conventions and closed flag walks

Human Compression project

The survey of Schulte–Weiss uses consecutive rotations, whereas our chamber arguments use rotations based at one fixed type. We give the change of generators and recover the survey's period formulas for holes and zigzags. The formulas are established results recorded in the survey; no new classification or answer to its Problem 7 is asserted.

1 Two choices of rotations

Let $n \geq 3$ and let a group G have generators $\sigma_1, \ldots, \sigma_{n-1}$ satisfying

$$(\sigma_i \sigma_{i+1} \cdots \sigma_j)^2 = 1 \qquad (1 \leq i < j \leq n-1).$$

Put $\alpha_0 = 1$ and $\alpha_i = \sigma_1 \cdots \sigma_i$. Then $\sigma_i = \alpha_{i-1}^{-1}\alpha_i$, and $\alpha_i^2 = 1$ for $i \geq 2$. These are identities in the marked group, independent of an assertion that its coset geometry is a polytope.

Proposition 1. *An automorphism θ simultaneously inverts every α_i if and only if*

$$\theta(\sigma_1) = \sigma_1^{-1}, \qquad \theta(\sigma_2) = \sigma_1^2\sigma_2, \qquad \theta(\sigma_i) = \sigma_i \quad (i \geq 3).$$

If such an automorphism exists, its square is the identity.

Proof. Suppose first that $\theta(\alpha_i) = \alpha_i^{-1}$. The assertion for $\sigma_1 = \alpha_1$ follows at once. Since α_2 is an involution,

$$\theta(\sigma_2) = \theta(\alpha_1^{-1}\alpha_2) = \alpha_1\alpha_2 = \sigma_1^2\sigma_2.$$

For $i \geq 3$ both α_{i-1} and α_i are involutions, so they are fixed by θ, and their product σ_i is fixed. Conversely the displayed formulas give $\theta(\alpha_1) = \alpha_1^{-1}$ and $\theta(\alpha_2) = \sigma_1^{-1}\sigma_1^2\sigma_2 = \alpha_2$. Multiplying by the fixed later generators gives $\theta(\alpha_i) = \alpha_i = \alpha_i^{-1}$ for every $i \geq 2$. Applying θ twice fixes the generating set (α_i). □

Thus the two reflection tests agree whenever the appropriate string-group reconstruction theorem applies. We retain that reconstruction theorem, including its intersection hypotheses, as an attributed input. A presentation with the displayed relations alone is not a reconstruction proof. Also θ may be the identity; in that case a geometric reflection interchanging two chamber orbits is still a separate involution, as in the accepted residue theorem.

2 Automorphisms and adjacency operations

For a polyhedron let r_i send a flag to its unique i-adjacent flag. The symbols r_i in this section denote permutations of flags. They need not be automorphisms of the polyhedron. They satisfy $r_i^2 = 1$ and $r_0 r_2 = r_2 r_0$. A word is applied in its written order; write Φ^w for its result on a flag Φ. Automorphisms commute with all these operations because adjacency is intrinsic.

Lemma 2. *Suppose the automorphism group acts freely and transitively on a set of flags $\mathcal{O}$, and a word w preserves $\mathcal{O}$. Let g_w be the unique automorphism with $g_w\Phi = \Phi^w$. Then $\Phi^{w^k} = g_w^k\Phi$. The least positive return time of this flag walk is the order of g_w, with both interpreted as infinite if there is no return.*

Proof. An automorphism fixing one flag fixes its adjacent flags and hence all flags by connectedness; thus the freeness hypothesis holds in every application here. Commutation gives $(g_w^k\Phi)^w = g_w^k(\Phi^w) = g_w^{k+1}\Phi$, proving the identity by induction. The final assertion is freeness. □

In a regular polyhedron $\mathcal{O}$ is the full flag set. In a chiral polyhedron there are two such orbits and every adjacency interchanges them. A word of even length preserves each orbit, and a word of odd length interchanges them. Accordingly the lemma applies to even words in the chiral case.

3 Holes and zigzags

A flag records a vertex, an incident edge and one incident face. Crossing the edge is the operation r_0. At the new vertex, the operation r_1 takes the next edge in the selected face, and r_2 changes the selected face while keeping the vertex and edge. Repeating these two local operations goes around the vertex figure.

Fix $1 \le j \le \lfloor q/2 \rfloor$ for a polyhedron of type $\{p, q\}$. The following words each advance by one edge of the indicated walk:

$$h_j = r_0 r_1 (r_2 r_1)^{j-1}, \qquad z_j = r_0 (r_1 r_2)^j.$$

Indeed after r_0, the first r_1 selects the first outgoing edge; each additional $r_2 r_1$ selects the next one. This gives the jth outgoing edge. In h_j the selected face remains on the same side of the walk, and the successive turns have the same sense. The final r_2 in $z_j = h_j r_2$ chooses the opposite incident face, so the next turn has the opposite sense. These are respectively the j-hole and j-zigzag rules. Their lengths are their complete traversal periods, including the selected side of the traversal, rather than the first return to a previously visited vertex.

For a regular polyhedron, let ρ_i be the distinguished automorphism with $\rho_i\Phi = \Phi^{r_i}$. Commutation with adjacency shows inductively that a word in the ρ_i has the same effect on Φ as the written word in the r_i. Lemma 2 therefore gives

$$\ell(h_j) = \operatorname{ord}\bigl(\rho_0\rho_1(\rho_2\rho_1)^{j-1}\bigr), \qquad \ell(z_j) = \operatorname{ord}\bigl(\rho_0(\rho_1\rho_2)^j\bigr).$$

For a chiral polyhedron choose the distinguished rotations by $\sigma_1\Phi = \Phi^{r_0 r_1}$ and $\sigma_2\Phi = \Phi^{r_1 r_2}$. These are consecutive rotations with the chosen base orientation. Write $s_1 = r_0 r_1$, $s_2 = r_1 r_2$ as formal words in involutions. Direct cancellation gives

$$h_j = s_1 s_2^{1-j}, \qquad r_0 s_2^j r_0 = s_1 s_2^{-j} s_1^{-1}, \qquad z_j^2 = s_1 s_2^{-j} s_1^{-1} s_2^j.$$

For $j = 1$, direct cancellation gives

$$(r_0 r_1)(r_2 r_1)(r_1 r_0) = r_0 r_1 r_2 r_0 = r_0 s_2 r_0.$$

Taking the jth power proves the middle identity. The first and last words are even. Applying the lemma on the base flag orbit proves

$$\ell(h_j) = \operatorname{ord}(\sigma_1 \sigma_2^{1-j}),$$
$$\ell(z_j) = 2\operatorname{ord}(\sigma_1 \sigma_2^{-j} \sigma_1^{-1} \sigma_2^j).$$

The factor two is necessary: one zigzag step exchanges the two flag orbits, so a return cannot have odd length. The alternative word $\sigma_2^j \sigma_1 \sigma_2^{-j} \sigma_1^{-1}$ is conjugate to the word displayed above, by σ_2^j, and has the same order.

These computations identify the exact flag periods. They neither say that the period data determine the polyhedron nor prove finiteness of a universal quotient with prescribed periods. Those are the separate questions posed in the survey.

References

[1] E. Schulte and A. Ivić Weiss, *Problems on polytopes, their groups, and realizations*, Period. Math. Hungar. 53 (2006), 231–255; fixed arXiv math/0608397v1, Sections 2–3. We use the survey's definitions and its explicit attribution of the reconstruction theorem to their earlier work on chiral polytopes.

Universal regular amalgamation from a witness

Human Compression project

We recover the conditional existence assertion in Section 4 of the fixed Schulte–Weiss survey. Its premise is that an amalgamation already exists; the argument does not decide which local pairs have an amalgamation. We retain the standard end criterion for a string C-group and the correspondence between such marked groups and regular polytopes. Both are the same attributed inputs used by our acquired chamber theory. The construction below is standard universal-presentation reasoning, written out to keep its existence premise visible.

Theorem 1. *Let K, L be regular polytopes of rank $n \geq 2$. Suppose a polytope P of rank $n+1$ has all facets isomorphic to K and all vertex figures isomorphic to L. Then there is a regular polytope U with those local types which covers every polytope with those local types, after choosing base flags. Its group is obtained from the universal string group by imposing all the marked facet and vertex-figure relations.*

Proof. Let

$$W = \langle s_0, \ldots, s_n \mid s_i^2 = 1, (s_i s_j)^2 = 1 \ (|i-j| > 1) \rangle.$$

For $I \subseteq \{0, \ldots, n\}$ write $W_I = \langle s_i : i \in I \rangle$. Put $A = \{0, \ldots, n-1\}$, $B = \{1, \ldots, n\}$ and $C = A \cap B$. The standard markings give onto homomorphisms $W_A \to \mathrm{Aut}(K)$ and $W_B \to \mathrm{Aut}(L)$, with kernels N_A, N_B. Let N be their normal closure in W, and put $G = W/N$. Use G_I for the subgroup on the indicated images of the marks.

The generators of W act by the corresponding adjacency operations on the flags of P. These operations are involutions, and nonconsecutive ones commute, so this is an action of W (with words applied in their written order). Every relation in N_A acts trivially on all flags: it stays inside a facet, where the adjacency action is the regular flag action of a copy of K. This assertion holds in every facet, since an isomorphism of regular polytopes can send the chosen flag to any flag. The same argument in vertex figures applies to N_B. Consequently the action factors through G.

Fix a base flag Φ of P. The action of G_A on the flags containing the base facet is precisely the regular flag action of K. More explicitly, the quotient map induces $W_A/N_A \twoheadrightarrow G_A$, while the restriction of the action back to that facet has kernel exactly N_A in W_A. These two observations show that $G_A \cong W_A/N_A$ as marked groups. In particular G_A is a C-group and its action on this facet's flags is free. Similarly $G_B \cong W_B/N_B$, and its action on the flags containing the base vertex is free.

We claim $G_A \cap G_B = G_C$. If g belongs to the intersection, then Φ^g has both the base vertex and the base facet. The flags containing these two faces form the flag set of their section. They are connected by adjacencies of types $1, \ldots, n-1$. There is therefore $h \in G_C$ with $\Phi^h = \Phi^g$. Both h and g lie in G_A, whose action on the facet is free, so $g = h$. The reverse inclusion is immediate. The end criterion now proves that G is a string C-group. Its marked regular polytope U has facets K and vertex figures L by the exact end-group identifications just proved.

For any other polytope Q with those local types, the same local argument shows that N acts trivially on its flags. Choose a base flag Ψ. Identify the flags of U with G in such a way that adjacency of type i is right multiplication by s_i. The map

$$g \longmapsto \Psi^g$$

is well-defined, onto, and commutes with each adjacency operation. It induces a rank-preserving incidence map on faces: flags containing a fixed face of rank i form an orbit under adjacencies of all types except i, and their images therefore contain one fixed rank-i face of Q. This gives a face map independent of the chosen incident flag. Images of a flag remain a flag, and adjacent flags remain adjacent. Thus it is a polytope covering. This proves universality. □

The proof needs no regularity assumption on the witness P. In particular it includes a chiral witness with regular facets and vertex figures, as in the survey. It uses regularity of the two local types essentially: their marked relations must act trivially on every local flag, and the local action used to identify the middle intersection must be free. The oriented amalgamation assertion for chiral constituents has different input data, including relative enantiomorphic choices; it is retained separately with its source attribution.

The group presentation describes the universal object when the premise is satisfied. It does not show that every pair with matching intermediate sections has a witness, does not supply an order algorithm, and does not imply finiteness. Thus the theorem is the established conditional statement, not a solution of the survey's amalgamation or finiteness questions.

References

[1] E. Schulte and A. Ivić Weiss, *Problems on polytopes, their groups, and realizations*, Period. Math. Hungar. 53 (2006), 231–255; fixed arXiv math/0608397v1, Section 4 and its attribution to P. McMullen and E. Schulte, *Abstract Regular Polytopes*. The standard end criterion and marked reconstruction are retained inputs, not new results of this note.

Two oriented square-torus amalgamations

Human Compression project

The fixed Schulte–Weiss survey attributes the following two examples to Hubard–Weiss, *Self-duality of chiral polytopes* (2005), p. 132. We verify their presented groups and the actual intersection and chirality conditions. This is a recovery of two established examples, not a new existence result or a classification of toroidal amalgamations.

Write $T(s,t) = \{4,4\}_{(s,t)}$ with its specified orientation, and put

$$R(x,y;s,t) = (x^{-1}y)^s(xy^{-1})^t.$$

We retain two precise standard interfaces stated in the survey: the oriented square-torus presentation on rotations x, y is $x^4 = y^4 = (xy)^2 = R(x,y;s,t) = 1$; and the rank-four rotation reconstruction criterion consists of the string rotation relations and the three intersection equalities below. The square-torus convention is the lattice spanned by (s,t) and $(-t,s)$. In the cases here its index is ten, so its rotation group has order forty. No completeness assertion for other parameters is used.

Proposition 1. *For $(s,t) = (1,3)$ or $(3,1)$, let*

$$G(s,t) = \langle u,v,w \mid u^4 = v^4 = w^4 = (uv)^2 = (vw)^2 = (uvw)^2 = 1,$$
$$R(u,v;s,t) = R(v,w;1,3) = 1\rangle.$$

These are the automorphism groups of the oriented universal chiral amalgamations with facets $T(s,t)$ and vertex figures $T(1,3)$. Their orders are respectively 2000 *and* 960*; hence the two resulting polytopes are not isomorphic.*

Proof. The supplied finite calculation constructs an isomorphism from each full finitely presented group to a permutation group, of degrees fifty and twenty respectively. This is not merely a finite quotient representation. The following quantities are checked in those actual permutation groups:

(s,t)	$\lvert G(s,t)\rvert$	$\lvert\langle u,v\rangle\rvert$	$\lvert\langle v,w\rangle\rvert$	$\lvert\langle u,v\rangle \cap \langle v,w\rangle\rvert$
$(1,3)$	2000	40	40	4
$(3,1)$	960	40	40	4

Each of u, v, w has order four, and

$$\langle u\rangle \cap \langle v\rangle = \langle v\rangle \cap \langle w\rangle = 1, \qquad \langle u,v\rangle \cap \langle v,w\rangle = \langle v\rangle.$$

The local finitely presented groups are separately checked to have order forty. Their natural epimorphisms onto the actual end subgroups are therefore isomorphisms, so the prescribed toroids do not collapse. The retained reconstruction criterion now supplies a rank-four polytope with precisely those oriented local types.

The regularity test would require an automorphism taking

$$(u,v,w) \longmapsto (u^{-1}, u^2v, w).$$

After this substitution, both displayed toroidal relators are nonidentity in each group. Such an automorphism cannot exist. The reconstructed polytopes are therefore chiral and $G(s,t)$ is their full automorphism group, rather than only the rotation subgroup of a regular polytope.

The two added relators are exactly the local defining relations beyond the standard rotation relations. Every oriented amalgamation with these specified facets and vertex figures consequently has its marked rotation group as a quotient of $G(s,t)$. Together with the verified existence and uncollapsed local types, the retained oriented amalgamation interface gives the stated universal object. Finally unequal automorphism-group orders exclude an isomorphism between the two polytopes. □

Verification and attribution. The executable calculation uses GAP's isomorphism algorithm for finite presentations. It checks bijectivity and all defining relations, and exports the three literal permutations. A separate Python calculation, using only composition and exhaustive closure of the resulting small groups, verifies both group orders, every stated subgroup intersection, local orders and both failed reflection relators. This second calculation does not itself prove the upper order of the finite presentation; that step remains the explicit GAP computation. A damaged marking is rejected. These are computational checks, not Lean certificates.

The source is the fixed arXiv edition math/0608397v1, Section 4 (physical p. 7) for the orders and attribution, and Section 5 (physical p. 12) for the literal toroidal relator. The rank-four reconstruction and reflection criteria occur in Section 2 (physical p. 4), attributed there to Schulte–Weiss, *Chirality and projective linear groups* (1994). The cited 1994 and 2005 primary papers were not independently inspected for this component; the exact interfaces above are retained from the read survey, while the finite group and intersection calculations are reproduced here. General chiral amalgamation existence or finiteness questions are not resolved by these two examples.